

A generalized Dunnett test for multi-arm multi-stage clinical studies with treatment selection

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SUMMARY

We generalize the Dunnett test to derive efficacy and futility boundaries for a flexible multi-arm multi-stage clinical trial for a normally distributed endpoint with known variance. We show that the boundaries control the familywise error rate in the strong sense. The method is applicable for any number of treatment arms, number of stages and number of patients per treatment per stage. It can be used for a wide variety of boundary types or rules derived from α -spending functions. Additionally, we show how sample size can be computed under a least favourable configuration power requirement and derive formulae for expected sample sizes.

Some key words: Clinical trial; Dunnett test; Familywise error rate; Treatment selection.

1. INTRODUCTION

The comparison of several distinct treatments with a common control occurs frequently in clinical trials. Often the distinct treatments are different doses of the same drug or different combinations of drugs. A range of well-established fixed-sample methods exist for hypothesis testing in this context (e.g., Hsu, 1996, Ch. 3). It may, however, be desirable to monitor the trial at a series of interim analyses to allow early stopping if efficacy is quickly established and/or to allow treatments that appear unlikely to have any benefit to be dropped early. A natural approach would be to combine fixed-sample multiple-comparison methods with group-sequential single-comparison designs (Jennison & Turnbull, 2000, Ch. 4). This has proved difficult without certain simplifying assumptions, such as only allowing one treatment to continue beyond the first interim (Thall et al., 1988) or, more generally, prespecifying the number of treatments at each stage (Stallard & Friede, 2008). Follmann et al. (1994) have considered the related problem of repeated two-sided multiple hypothesis testing, where treatments are only dropped at interim if found significantly inferior to control. Sometimes, when resources are limited and there is a pressing need to find at least one effective treatment, this requirement is too stringent.

In this paper, we show how efficacy and futility boundaries can efficiently be computed for a multi-arm multi-stage trial with a normally distributed endpoint. The number of treatment arms within any stage is not prespecified and we allow a general number of patients per treatment per stage. We explore the case in which it is desirable to stop the study as soon as it is clear that either no treatments are superior to the control or at least one of them is significantly superior. Boundaries such as those due to Pocock (1977) or O'Brien & Fleming (1979) can be accommodated, as can the triangular test (Whitehead, 1997) or an α -spending approach (Lan & DeMets, 1983). Our approach generalizes the one-stage Dunnett test (Dunnett, 1955) and we prove control of the familywise error rate in the strong sense, defined as the probability of rejecting one or more true null hypotheses. After imposing appropriate constraints, the boundaries are computed via numerical integration and a one-dimensional search. We describe how sample size may subsequently be found in a separate one-dimensional search by consideration of the least favourable configuration of treatment effects. Finally, we derive the distribution for the number of stages a treatment is included in the study which in turn enables exact computations for expected sample sizes for any parameter configuration.

The main advantage of using multi-arm, multi-stage designs lies in the efficient evaluation of several treatments. The sample size and duration required tends to be markedly smaller than if each treatment had been evaluated separately (e.g., [Barthel et al., 2009](#)). One trial that could have benefited greatly from such a design was the comparison of three different doses of riluzole with placebo to treat a serious neurodegenerative disorder for which there was no existing therapy ([Lacomblez et al., 1996](#)). In this case, clarification of at least one effective dose in a timely and efficient manner was of clear importance. Despite the advantages of this approach, only a very limited number of such trials have been undertaken (e.g., [Sydes et al., 2009](#)) due to a lack of available methods.

2. MODEL AND HYPOTHESIS TESTING

2.1. Familywise error rate

Let $\mu_1, \dots, \mu_K$ be the mean responses on K experimental treatments and μ_0 be the mean response on the control. The null hypotheses of interest are $H_1: \mu_1 \leq \mu_0, \dots, H_K: \mu_K \leq \mu_0$ and it is commonly required that the probability of one or more false rejections within this family be no greater than α . This is known as strong control of the familywise error rate. We assume that each patient, independently, provides a normally distributed response with known variance, σ^2 . Ideas for dealing with an unknown variance are outlined in the discussion.

As data accumulate over the course of the trial, the family of null hypotheses is tested at each of a series of analyses, indexed by $j = 1, \dots, J$. At analysis j , to test H_k , it is assumed that responses, $X_{q,i}$, from patients $i = 1, \dots, r_{q,j}n$ on treatments $q = k, 0$ are available. The notation $r_{q,j}$ for the ratios of sample sizes is convenient, as it will be shown that specification of appropriate stopping boundaries does not depend on the absolute sample sizes. The test statistics $Z_{k,j} = (\hat{\mu}_{k,j} - \hat{\mu}_{0,j})I_{k,j}^{1/2}$ are used, where $I_{k,j} = r_{k,j}r_{0,j}n\{\sigma^2(r_{k,j} + r_{0,j})\}^{-1}$ and $\hat{\mu}_{q,j} = (r_{q,j}n)^{-1} \sum_{i=1}^{r_{q,j}n} X_{q,i}$ ($q = 0, k$). Upper and lower stopping boundaries, $U = (u_1, \dots, u_J)$ and $L = (l_1, \dots, l_J)$, are defined as follows. If $Z_{k,j} > u_j$, then H_k is rejected and the whole study is stopped with the conclusion that the k th experimental treatment is superior to control. If $Z_{k,j} < l_j$, treatment k is dropped from all subsequent stages with $Z_{k,l} = -\infty$ ($l = j + 1, \dots, J$). Otherwise, treatment k continues into stage $j + 1$. Note that $u_J = l_J$.

For any choice of μ_k and $r_{k,j}$ ($k = 0, \dots, K$; $j = 1, \dots, J$), the probability of rejecting at least one null hypothesis can be found by integrating the joint density of the test statistics, which is assumed to be multivariate normal with known correlation matrix, over some region of $K \times J$ -dimensional space. Direct numerical integration becomes difficult as both the number of stages and the number of treatments increase. However, following the method of [Dunnnett \(1955\)](#) for a single stage trial, one can exploit the fact that correlation between test statistics for different experimental treatments arises solely from the common control responses. In other words, for $k \neq k'$, the statistics $Z_{k,1}, \dots, Z_{k,J}$ are conditionally independent of $Z_{k',1}, \dots, Z_{k',J}$ given $\hat{\mu}_{0,1}, \dots, \hat{\mu}_{0,J}$.

For any vector of constants, $\Theta = (\theta_1, \dots, \theta_K)$ and $k = 1, \dots, K$, $j = 1, \dots, J$ define the events

$$A_{k,j}(\theta_k) = \{Z_{k,j} < l_j + (\mu_k - \mu_0 - \theta_k)I_{k,j}^{1/2}\},$$

$$B_{k,j}(\theta_k) = \{l_j + (\mu_k - \mu_0 - \theta_k)I_{k,j}^{1/2} < Z_{k,j} < u_j + (\mu_k - \mu_0 - \theta_k)I_{k,j}^{1/2}\}.$$

If $\mu_k - \mu_0 = \theta_k$ ($k = 1, \dots, K$), the event that $H_1, \dots, H_K$ all fail to be rejected is equivalent to

$$\bar{R}_K(\Theta) = \bigcap_{k=1}^K \left(\bigcup_{j=1}^J \left[\left\{ \bigcap_{i=1}^{j-1} B_{k,i}(\theta_k) \right\} \cap A_{k,j}(\theta_k) \right] \right),$$

with the convention that $\bigcap_{i=1}^0 = \Omega$, where Ω is the whole sample space.

Rearranging, $A_{k,j}(\theta_k) = \{(\hat{\mu}_{k,j} - \mu_k)\sigma^{-1}(r_{k,j}n)^{1/2} < l_{k,j}(\theta_k)\}$ with

$$l_{k,j}(\theta_k) = \frac{r_{k,j}^{1/2}}{r_{0,j}} \sum_{i=1}^j (r_{0,i} - r_{0,i-1})^{1/2} t_i - \frac{\theta_k}{\sigma} (r_{k,j}n)^{1/2} + l_j \lambda_{k,j}, \quad (1)$$

where $\lambda_{k,j} = (1 + r_{k,j}/r_{0,j})^{1/2}$ and $r_{0,0} = 0$. Also,

$$t_i = \{n(r_{0,i} - r_{0,i-1})\}^{-1/2} \sum_{l=1+nr_{0,i-1}}^{nr_{0,i}} (X_{0,l} - \mu_0)/\sigma \quad (i = 1, \dots, j),$$

are independent standard normal. Similarly, $B_{k,j}(\theta_k) = \{l_{k,j}(\theta_k) < (\hat{\mu}_{k,j} - \mu_k)\sigma^{-1}(r_{k,j}n)^{1/2} < u_{k,j}(\theta_k)\}$, with $u_{k,j}(\theta_k) = l_{k,j}(\theta_k) + (u_j - l_j)\lambda_{k,j}$. Thus,

$$\begin{aligned} \text{pr}\{\bar{R}_K(\Theta)\} &= E(\cdots E[\text{pr}\{\bar{R}_K(\Theta)\} | t_J] \cdots | t_1) \\ &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \prod_{k=1}^K \left[\sum_{j=1}^J \Phi_j\{L_{k,j}(\theta_k), U_{k,j}(\theta_k), \Sigma_{k,j}\} \right] d\Phi(t_1) \cdots d\Phi(t_J), \end{aligned} \quad (2)$$

where Φ denotes the standard normal distribution function and $\Phi_j\{L_{k,j}(\theta_k), U_{k,j}(\theta_k), \Sigma_{k,j}\}$ denotes the result of integrating the j -dimensional normal density with mean zero and correlation matrix $\Sigma_{k,j}$ over a region defined by a vector of lower limits, $L_{k,j}(\theta_k) = \{l_{k,1}(\theta_k), \dots, l_{k,j-1}(\theta_k), -\infty\}$, and a corresponding vector of upper limits, $U_{k,j}(\theta_k) = \{u_{k,1}(\theta_k), \dots, u_{k,j-1}(\theta_k), l_{k,j}(\theta_k)\}$. The (i, i') th element ($i < i'$) of $\Sigma_{k,j}$ is $(r_{k,i}/r_{k,i'})^{1/2}$.

THEOREM 1. Denote the global null hypothesis, $\mu_0 = \mu_1 = \cdots = \mu_K$, by H_G . For any Θ , $\text{pr}(\text{reject at least one true } H_k | \Theta) \leq \text{pr}(\text{reject at least one } H_k | H_G)$.

Proof. It is shown in the Appendix that for any $\epsilon_k > 0$,

$$\bigcup_{j=1}^J \left[\left\{ \bigcap_{i=1}^{j-1} B_{k,i}(\theta_k + \epsilon_k) \right\} \cap A_{k,j}(\theta_k + \epsilon_k) \right] \subseteq \bigcup_{j=1}^J \left[\left\{ \bigcap_{i=1}^{j-1} B_{k,i}(\theta_k) \right\} \cap A_{k,j}(\theta_k) \right].$$

Next suppose without loss of generality that $\theta_1, \dots, \theta_l \leq 0$ and $\theta_{l+1}, \dots, \theta_K > 0$. Let $\Theta_l = (\theta_1, \dots, \theta_l)$. Then,

$$\begin{aligned} \text{pr}(\text{reject at least one true } H_k | \Theta) &\leq \text{pr}[Z_{k,j} > u_j \text{ for some } (k, j) \in \{1, \dots, l\} \times \{1, \dots, J\} | \Theta] \\ &= 1 - \text{pr}\{\bar{R}_l(\Theta_l)\} \\ &\leq 1 - \text{pr}\{\bar{R}_l(0)\} \\ &\leq 1 - \text{pr}\{\bar{R}_K(0)\} \\ &= \text{pr}[Z_{k,j} > u_j \text{ for some } (k, j) \in \{1, \dots, K\} \times \{1, \dots, J\} | H_G] \\ &= \text{pr}(\text{reject at least one } H_k | H_G). \end{aligned} \quad \square$$

The right-hand side of (2) can be evaluated numerically for any choice of $n, r_{k,j}, u_j$ and l_j ($j = 1, \dots, J$; $k = 0, \dots, K$) with $l_J = u_J$. However, it is clear from (1) that if $\Theta = 0$ this probability is the same for all choices of n . Therefore, setting $\text{pr}(\text{reject at least one } H_k | H_G) = \alpha$, i.e., $\text{pr}\{\bar{R}_K(0)\} = 1 - \alpha$, results in an equation with $J(K + 1) + 2J - 1$ unknowns and hence practical constraints must be used to solve it. By Theorem 1, the familywise error rate is controlled in the strong sense.

2.2. Constraints

Consider the special case of a single stage design, resulting in a single critical value, $u_1 = l_1$, for testing all hypotheses H_k . Under H_G , (2) reduces to $\text{pr}\{\bar{R}_K(0)\} = \int_{-\infty}^{\infty} \prod_{k=1}^K \Phi\{t_1(\lambda_{k,1}^2 - 1)^{1/2} + u_1\lambda_{k,1}\} d\Phi(t_1)$. After specifying the allocation ratios, numerical integration and a search can be used to find u_1 . This corresponds to the [Dunnett \(1955\)](#) test.

In the general case, a useful constraint is to have equal numbers of patients on each experimental treatment within each stage: $r_{1,j} = \dots = r_{K,j} = r_j$ ($j = 1, \dots, J$). A different number of patients may be placed on the control arm. For any constant, c , this gives $L_{1,j}(c) = \dots = L_{K,j}(c) = L_j(c)$, $U_{1,j}(c) = \dots = U_{K,j}(c) = U_j(c)$ and $\Sigma_{1,j} = \dots = \Sigma_{K,j} = \Sigma_j$, simplifying equation (2):

$$\text{pr}\{\bar{R}_K(0)\} = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \left[\sum_{j=1}^J \Phi_j\{L_j(0), U_j(0), \Sigma_j\} \right]^K d\Phi(t_1) \dots d\Phi(t_J). \quad (3)$$

Equation (3) may be used when designing the trial. A sensible choice must be made for the r_j and $r_{0,j}$, e.g., $r_j = r_{0,j} = jr_1$ or $r_j = r_{0,j} = r_1(j+1)/2$. As the interim analyses are conducted, boundaries may be adjusted using (2) to account for the number of responses actually observed.

This leaves an equation with $2J - 1$ unknowns, $u_1, \dots, u_J$ and $l_1, \dots, l_{J-1}$. One option is to use a multi-arm analogue of a [Pocock \(1977\)](#) design for the efficacy boundary with $u_1 = \dots = u_J = a$. Using a symmetric futility boundary, $l_j = -u_j$, may not be very helpful when considering one-sided tests as this would be too stringent a requirement for dropping ineffective treatments. A simple alternative is to put $l_j = 0$ ($j = 1, \dots, J - 1$), i.e., an experimental treatment is dropped whenever it is performing at all worse than control at interim. This is intuitively appealing, although more sophisticated approaches could be taken such as using conditional power. Similarly, a multi-arm [O'Brien & Fleming \(1979\)](#) design analogue could be used for the efficacy boundary with $u_j = a(r_j/r_J)^{1/2}$, or a triangular test ([Whitehead, 2011](#)) analogue with $l_j = -a(1 - 3r_j/r_J)/r_j^{1/2}$ and $u_j = a(1 + r_j/r_J)/r_j^{1/2}$. This leaves one unknown and one equation.

An alternative is to employ an α -spending function approach ([Lan & DeMets, 1983](#)) making repeated use of (2) to iteratively compute the efficacy stopping boundary for a given futility boundary. That is, at the j th interim, one solves

$$\alpha^*(\tau_j) = 1 - \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{k=1}^K \left[\sum_{i=1}^j \Phi_i\{L_{k,i}(0), U_{k,i}(0), \Sigma_{k,i}\} \right] d\Phi(t_1) \dots d\Phi(t_j)$$

for u_j , given $(u_1, \dots, u_{j-1})$ from previous interims and a prespecified nondecreasing α -spending function, $\alpha^*(\tau)$, with $\alpha^*(0) = 0$ and $\alpha^*(1) = \alpha$. This requires some definition of information time, $0 < \tau < 1$, in the multi-arm setting ([Follmann et al., 1994](#)).

3. POWER AND SAMPLE SIZE

Following the approach of [Dunnett \(1984\)](#), the power of the study can be defined by consideration of two quantities: δ , an effect size such that, if $\mu_k - \mu_0 > \delta$, we would wish to proceed further with treatment k , i.e., δ represents a clinically relevant improvement; and an effect size δ_0 such that, if $\mu_k - \mu_0 < \delta_0$, then we would prefer not to proceed further in investigating treatment k . It may be that δ_0 is set equal to zero, implying that any advantage would be worth detecting, but more likely δ_0 represents a marginal improvement that is not worth spending resources to establish. For the purposes of defining power, it is assumed that the treatment recommended at the end of the study corresponds to the first test statistic to cross the efficacy boundary or, should two or more test statistics cross at once, to the largest of these test statistics. It is also assumed that sample sizes are equal for the experimental treatments within each stage. Power can now be defined as follows: the probability that, without loss of generality, H_1 is rejected and treatment 1 recommended given $\mu_1 - \mu_0 = \delta$ and $\mu_k - \mu_0 = \delta_0$ ($k = 2, \dots, K$). This is known as the least favourable configuration because it minimizes the probability of recommending treatment 1 over all choices of $\mu_0, \dots, \mu_K$ such that $\mu_1 - \mu_0 \geq \delta$ and $\mu_k - \mu_0 \leq \delta_0$ ($k = 2, \dots, K$) ([Thall et al., 1988](#)).

Denote the probability under the least favourable configuration that no individual null hypotheses are rejected at analyses $1, \dots, J-1$ and then at analysis J , H_1 is rejected and treatment 1 is recommended by Π_J . The power of the study is then $\Pi_1 + \Pi_2 + \dots + \Pi_J$.

As in § 2, numerical evaluation of Π_J can be simplified by taking advantage of conditional independence. Here it is $\hat{\mu}_{1,J}$ rather than $\hat{\mu}_{0,J}$ that is common to all comparisons of interest at the J th analysis. After conditioning over $\hat{\mu}_{0,1}, \dots, \hat{\mu}_{0,J-1}$ and $\hat{\mu}_{1,J}$, some elementary manipulation provided in the Supplementary Material yields

$$\Pi_J = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left[\sum_{j=1}^J \Phi_j \{L_j(\delta_0), U_j(\delta_0), \Sigma_j\} \right]^{K-1} \\ \times \Phi_{J-1} \{\bar{L}_{J-1}(\delta), \bar{U}_{J-1}(\delta), \bar{\Sigma}_{J-1}\} \Phi(W_J) d\Phi(t_1) \cdots d\Phi(t_{J-1}) d\Phi(v_J), \quad (4)$$

where Φ , Φ_j , $L_j(\delta_0)$, $U_j(\delta_0)$ and Σ_j are defined as in § 2.2. Also, $l_j(\cdot)$ and $u_j(\cdot)$ are defined as before for $j = 1, \dots, J-1$ with $l_J(\delta_0) = v_J + (\delta - \delta_0)\sigma^{-1}(r_J n)^{1/2}$. Additionally, $\bar{L}_{J-1}(\delta) = \{\bar{l}_1(\delta), \dots, \bar{l}_{J-1}(\delta)\}$ and $\bar{U}_{J-1}(\delta) = \{\bar{u}_1(\delta), \dots, \bar{u}_{J-1}(\delta)\}$ with $\bar{l}_j(\delta) = \{r_J/(r_J - r_j)\}^{1/2} \{l_j(\delta) - (r_j/r_J)^{1/2} v_J\}$ and $\bar{u}_j(\delta) = \{r_J/(r_J - r_j)\}^{1/2} \{u_j(\delta) - (r_j/r_J)^{1/2} v_J\}$. Also, $\bar{\Sigma}_{J-1}$ is a symmetric $(J-1) \times (J-1)$ -dimensional matrix with (i, j) th entry given by $[r_i(r_J - r_j)]/\{r_j(r_J - r_i)\}^{1/2}$ for $i \leq j$ and, finally, $W_J = r_{0,J}\{r_J(r_{0,J} - r_{0,J-1})\}^{-1/2}\{v_J - u_J(\delta)\} - t_J$, with $u_J(\delta)$ and t_J defined as in § 2.

In particular, for a one-stage trial the power is Π_1 and (4) reduces to

$$\Pi_1 = \int_{-\infty}^{\infty} \left[\Phi \left\{ v_1 + \frac{(\delta - \delta_0)}{\sigma} (r_1 n)^{1/2} \right\} \right]^{K-1} \Phi \left[\left(\frac{r_{0,1}}{r_1} \right)^{1/2} \left\{ v_1 + \frac{\delta}{\sigma} (r_1 n)^{1/2} - u_1 \lambda_1 \right\} \right] d\Phi(v_1).$$

For a J -stage trial, power can be specified at $1 - \beta$ and the following equation solved numerically for n : $\Pi_1 + \dots + \Pi_J = 1 - \beta$.

4. EXPECTED SAMPLE SIZE

Expected sample size can be computed for a given design specified by L , U , n and $r_{k,j}$, ($k = 0, 1, \dots, K$; $j = 1, \dots, J$). Assuming that $\mu_k - \mu_0 = \theta_k$, the joint distribution of the number of stages, S_k , that treatment k is included in the trial ($k = 1, \dots, K$) is denoted by $f(s_1, \dots, s_K | \Theta) = \text{pr}(S_1 = s_1, \dots, S_K = s_K | \Theta)$, where $s_1, \dots, s_K \in \{1, \dots, J\}$. For the event $\bigcap_k S_k = s_k$ to occur, it is necessary that $l_j < Z_{k,j} < u_j$ ($j = 1, \dots, s_k - 1$; $k = 1, \dots, K$) and, after defining $m = \max_{k=1, \dots, K} s_k$, whenever $s_k < m$ it is also necessary that $Z_{k,s_k} < l_{s_k}$. Either all remaining treatments are dropped at stage m or the trial is stopped due to rejection of at least one null hypothesis at stage m . From the discussion in § 2.1, there is conditional independence of the $Z_{k,1}, \dots, Z_{k,J}$ and $Z_{k',1}, \dots, Z_{k',J}$ ($k \neq k'$) given $t_1, \dots, t_J$ and therefore,

$$f(s_1, \dots, s_K | \Theta) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left[\prod_{k: s_k < m} \Phi_{s_k} \{L_{k,s_k}(\theta_k), U_{k,s_k}(\theta_k), \Sigma_{k,s_k}\} \right] \\ \times \left(\left[\prod_{k: s_k = m} \Phi_m \{L_{k,m}(\theta_k), U_{k,m}(\theta_k), \Sigma_{k,m}\} \right] \right. \\ \left. + \left[\prod_{k: s_k = m} \Phi_{m-1} \{\tilde{L}_{k,m-1}(\theta_k), \tilde{U}_{k,m-1}(\theta_k), \Sigma_{k,m-1}\} \right] \right. \\ \left. - \left[\prod_{k: s_k = m} \Phi_m \{L_{k,m}(\theta_k), \tilde{U}_{k,m}(\theta_k), \Sigma_{k,m}\} \right] \right) d\Phi(t_1) \cdots d\Phi(t_m), \quad (5)$$

Table 1. *Stopping boundaries for a trial comparing four experimental treatments with control in either two or three stages based on three forms of constraints*

Design		L	U	n	$\min \mathcal{N}$	$\max \mathcal{N}$	$E(\mathcal{N} H_G)$	$E(\mathcal{N} \text{LFC})$
Two-stage	OBF	(0, 2.169)	(3.068, 2.169)	44	220	440	342.3	346.9
	P	(0, 2.375)	(2.375, 2.375)	50	250	500	382.9	318.9
	T	(0.811, 2.293)	(2.432, 2.293)	50	250	500	309.4	302.0
Three-stage	OBF	(0, 0, 2.182)	(3.779, 2.672, 2.182)	31	155	465	307.6	317.3
	P	(0, 0, 2.481)	(2.481, 2.481, 2.481)	37	185	555	359.2	286.3
	T	(0, 1.435, 2.343)	(2.706, 2.392, 2.343)	36	180	540	292.5	285.2

There are equal sample sizes per arm per stage (n) and minimum, maximum and expected total sample size are given under the global null hypothesis and the least favourable configuration. Boundaries and sample size were calculated for $\alpha = 0.05$ and $\beta = 0.1$. OBF, O'Brien and Fleming efficacy boundary with zero futility boundary; P, Pocock efficacy boundary with zero futility boundary; T, triangular test boundaries; LFC, least favourable configuration.

where ϕ , Φ_j , $L_{k,j}(\theta_k)$, $U_{k,j}(\theta_k)$, $\Sigma_{k,j}$ are defined as in § 2.1. Further, $\tilde{L}_{k,j}(\theta_k) = \{l_{k,1}(\theta_k), \dots, l_{k,j}(\theta_k)\}$ and $\tilde{U}_{k,j}(\theta_k) = \{u_{k,1}(\theta_k), \dots, u_{k,j}(\theta_k)\}$.

Equation (5) can be evaluated numerically for any argument of $f(\cdot | \Theta)$. Denoting the total sample size of a trial as $\mathcal{N}$, the expected sample size for given Θ is

$$E(\mathcal{N} | \Theta) = \sum_{s_1=1}^J \cdots \sum_{s_K=1}^J n f(s_1, \dots, s_K | \Theta) \left(r_{0,m} + \sum_{k=1}^K r_{k,s_k} \right).$$

5. EXAMPLE

We consider designing a trial comparing four experimental treatments with control. The clinically relevant improvement on experimental treatment is specified by $\text{pr}(X_{k,i} > X_{0,i}) = 0.65$. That is, the probability that a randomly chosen patient on treatment k does better than a randomly chosen patient on control is 0.65. As $\text{pr}(X_{k,i} > X_{0,i}) = \Phi\{2^{-1/2}(\mu_k - \mu_0)/\sigma\}$, this corresponds to $\delta = 2^{1/2}\sigma\Phi^{-1}(0.65)$. A marginal improvement that would not be worthwhile establishing is specified by $\text{pr}(X_{k,i} > X_{0,i}) = 0.55$, or, $\delta_0 = 2^{1/2}\sigma\Phi^{-1}(0.55)$.

Specifying treatment effects in this way has the advantage that an anticipated standard deviation is not required in equations (3)–(5), as σ will cancel. In practice, it is likely that the anticipated σ will nevertheless influence the choice of such probabilities.

Table 1 gives stopping boundaries, sample size per arm per stage, minimum total sample size, maximum total sample size and expected total sample size under the global null hypothesis and least favourable configuration for two-stage and three-stage versions of three designs described in § 2.2. Computations were carried out in R (R Development Core Team, 2012). The multivariate normal probabilities are calculated using the package mvtnorm (Genz et al., 2010) and the outer integrals in (2), (4) and (5) are evaluated using quadrature rules. For $J \geq 5$, we find quadrature becomes infeasible but Monte Carlo integration can be used. Code is available from the authors upon request.

Table 1 shows that the O'Brien–Fleming type design is preferable in terms of minimizing the minimum and maximum total sample size, whilst a triangular design is preferable in terms of minimizing the expected sample size under both the null hypothesis and the least favourable configuration. A three-stage design reduces the expected sample size compared with a two-stage design but at the cost of increasing the maximum sample size.

6. DISCUSSION

Our method of computing stopping boundaries avoids direct numerical integration of the joint density of the test statistics so that the computational cost of solving (3) and (4) increases only with the number of

stages and not with the number of treatments. Thus, it is very efficient for trials of two or three stages with a moderate number of treatments as there exist sophisticated algorithms for efficient numerical integration of bivariate and trivariate normal densities (e.g., [Genz & Bretz, 2009](#)).

We have assumed that the trial is stopped immediately upon any null hypothesis being rejected. It is possible to continue the trial and test the remaining null hypotheses whilst controlling the familywise error rate. Theorem 1 shows that the probability of rejecting at least one true null hypotheses throughout the J stages is at most α . Stopping the trial early is therefore conservative. If response is delayed and the trial stopped early this will lead to some loss of efficiency as responses will continue to accrue after termination. The problem of incorporating this additional data has been discussed in the context of single-comparison group-sequential trials in [Sooriyarachchi et al. \(2003\)](#).

The results are easily generalizable to binary or ordinal data following [Whitehead & Jaki \(2009\)](#) and the model may be extended to include covariates. Throughout this work, it has been assumed that the common variance is known. In large sample sizes, the effect of this assumption will become negligible. Explicitly incorporating this uncertainty would add considerable complexity to the approach described in this paper. In small sample sizes, a possible ad hoc adjustment would be to transform the individual test statistics using the function $f(x) = \Phi^{-1}\{T_\nu(x)\}$, where T denotes the cumulative distribution function of the t distribution with ν degrees of freedom for some suitably chosen ν . Details of this approach are given in [Jennison & Turnbull \(2000\)](#).

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SUPPLEMENTARY MATERIAL

Supplementary material available at *Biometrika* online includes the derivation of equation (4).

APPENDIX

Take any $\omega = (Z_{k,1}, \dots, Z_{k,J}) \in \bigcup_{j=1}^J [\{\bigcap_{i=1}^{j-1} B_{k,i}(\theta_k + \epsilon_k)\} \cap A_{k,j}(\theta_k + \epsilon_k)]$. For some $m \in \{1, \dots, J\}$, $Z_{k,m} \in A_{k,m}(\theta_k + \epsilon_k)$ and $Z_{k,j} \in B_{k,j}(\theta_k + \epsilon_k)$ for $j = 1, \dots, m-1$. But, $Z_{k,m} \in A_{k,m}(\theta_k + \epsilon_k)$ implies that $Z_{k,m} \in A_{k,m}(\theta_k)$ and $Z_{k,j} \in B_{k,j}(\theta_k + \epsilon_k)$ implies that $Z_{k,j} \in B_{k,j}(\theta_k) \cup A_{k,j}(\theta_k)$ for $j = 1, \dots, m-1$. Therefore, $\omega \in \bigcup_{j=1}^J [\{\bigcap_{i=1}^{j-1} B_{k,i}(\theta_k)\} \cap A_{k,j}(\theta_k)]$.

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